The documentation and process conversion measures necessary to comply with this revision shall be completed by 7 Oct 95.

INCH-POUND

MIL-S-19500/600A
7 April 1995
SUPERSEDING
MIL-S-19500/600
10 December 1993

MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, FIELD EFFECT TRANSISTOR, N-CHANNEL, SILICON
TYPES 2N7229 AND 2N7230
JAN, JANTX, JANTXV, JANS, JANHC, AND JANKC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for a N-Channel, enhancement-mode, MOSFET, power transistor intended for use in high density power switching applications. Four levels of product assurance are provided for each encapsulated device type as specified in MIL-S-19500, with avalanche energy ratings (E_{AS} and E_{AR}) and maximum avalanche current (I_{AS}). Two levels of product assurance are provided for each unencapsulated device type.

1.2 Physical dimensions. See figure 1, TO-254AA, and figure 2 and 6.3 for JANHC and JANKC die dimensions.

1.3 Maximum ratings. Unless otherwise specified, T_A = +25°C.

Type	V_{DSS} V_{GS} = 0 V dc I_D = 1.0 mA dc	P_T 1/ T_C = +25°C	V_{GS}	I_{D1} 2/ T_C = +25°C	I_{D2} 2/ T_C = +100°C	I_S	E_{AS}	E_{AR}	I_{AS} 2/	I_{DM} 3/
	V dc	W	V dc	A dc	A dc	A dc	mj	mj	A dc	A(pk)
2N7229	400	250	±20	23.0	14.0	23.0	500	25	23	92
2N7230	500	250	±20	19.0	12.0	19.0	500	25	19	76

Type	T_J and T_{STG}	V_{ISO} 70,000 feet altitued	MAX $R_{DS(ON)}$ 1/ V_{GS} = 10 V dc; I_D = I_{D2}		$R_{\theta JC}$ max
			T_J = +25°C	T_J = +150°C	
	°C		ohm	ohm	°C/W
2N7229	-55 to +150	400	0.200	0.495	0.50
2N7230	-55 to +150	500	0.270	0.665	0.50

1/ Derate linearly 2.0 W/°C for T_C > +25°C. $P_T = \dfrac{T_{J\,max} - T_C}{R_{\theta JC}}$

2/ $I_D = \sqrt{\dfrac{T_{J\,max} - T_C}{(R_{\theta JC} \times R_{DS(on)} \text{ at } T_{J\,max})}}$

3/ I_D = 4 I_{D1}; I_{D1} as calculated in footnote 2.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Electronics Supply Center, ATTN: DESC-ELDT, 1507 Wilmington Pike, Dayton, OH 454444-5270, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 5961

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1.4 Primary electrical characteristics at T_C = +25°C.

Type	Min $V_{(BR)DSS}$ V_{GS} = 0 V dc I_D = 1.0 mA dc	$V_{GS(th)1}$ $V_{DS} \geq V_{GS}$ I_D = 0.25 mA dc		Max I_{DSS1} V_{GS} = 0 V dc V_{DS} = 80 percent of rated V_{DS}	Max $r_{DS(on)}$ 1/ V_{GS} = 10 V dc; I_D = I_{D2} T_J = +25°C	T_J = +150°C	$R_{\theta JC}$ maximum
	V dc	V dc Min	Max	µA dc	ohm	ohm	°C/W
2N7229	400	2.0	4.0	25	0.200	0.495	0.50
2N7230	500	2.0	4.0	25	0.270	0.665	0.50

1/ Pulsed (see 4.5.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.4).

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

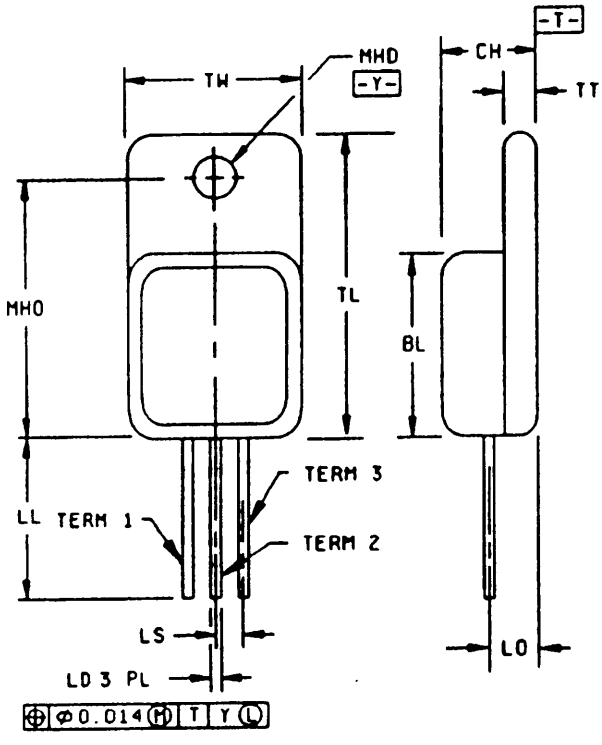
(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Defense Printing Service Detachment Office, Building. 4D (Customer Service), 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Associated detail specification. The individual item requirements shall be in accordance with MIL-S-19500 and as specified herein.

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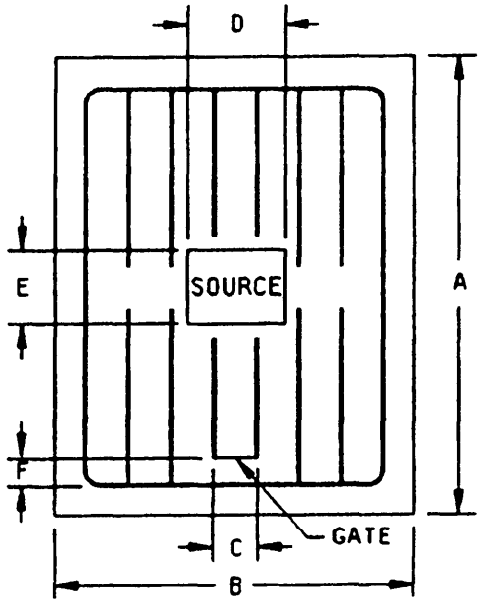


Symbol	Dimensions			
	Millimeters		Inches	
	Min	Max	Min	Max
CH	6.32	6.60	.249	.260
TT	1.02	1.27	.040	.050
LO	3.81		.150 BSC	
TL	20.07	20.32	.790	.800
BL	13.59	13.89	.535	.545
LS	3.81		.150 BSC	
TW	13.59	13.89	.535	.545
LL	13.59	13.89	1.195	1.235
MHO	16.89	17.40	.665	.685
MHD	1.53	3.78	.139	.149
LD	0.89	1.14	.035	.045
Term 1	Drain			
Term 2	Source			
Term 3	Gate			

NOTES:
1. Dimensions are in millimeters.
2. Inch equivalents are given for general information only.
3. Refer to applicable symbol list.
4. All terminals are isolated from case.
5. The preferred measurements used herein are the metric units. However, this transistor was designed using inch-pound units of measurement. In case of conflicts between the metric and inch-pound units, the inch-pound units shall be rule.
6. In accordance with ANSI Y14.5M, diameters are equivalent to φx symbology.

FIGURE 1. Dimensions and configuration (TO-254AA).

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Symbol	Dimensions			
	Millimeters		Inches	
	Min	Max	Min	Max
A	9.01	9.27	.355	.365
B	6.40	6.65	.252	.262
C	0.61	0.86	.024	.034
D	1.47	1.72	.058	.068
E	1.19	1.45	.047	.057
F	0.46	0.71	.018	.028

NOTES:
1. Dimensions are in millimeters.
2. Inch equivalents are given for general information only.
3. Unless otherwise specified, tolerance is 0.13 mm (±.0050 inch).
4. The physical characteristics of the die are: The back metals are chromium, nickel, and silver. The top metal is aluminum and the back contact is the drain.
5. The thickness is 0.474 mm (.01870 inch), the tolerance is 0.13 mm (±.0050 inch).

FIGURE 2. JANHC or JANKC die dimensions.

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3.2 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-S-19500.

3.3 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-S-19500 and on figures 1 and 2 herein. Methods used for electrical isolation of the terminal feedthroughs shall employ materials that contain a minimum of 90 percent AL_2O_3 (ceramic). Examples of such construction techniques are metallized ceramic eyelets or ceramic walled packages. The preferred measurements used herein are the metric units. However, this transistor was designed using inch-pound units of measurement. In case of conflicts between the metric and inch-pound units, the inch-pound units shall be rule.

3.3.1 Lead formation and finish. Lead finish shall be solderable in accordance with MIL-S-19500, MIL-STD-750, and herein.

3.3.2 Internal construction. Multiple chip construction shall not be permitted.

3.4 Marking. Marking shall be in accordance with MIL-S-19500.

3.5 Electrostatic discharge protection. The devices covered by this specification require electrostatic protection (see 6.2).

3.5.1 Handling. MOS devices must be handled with certain precautions to avoid damage due to the accumulation of static charge. The following handling procedures shall be followed:

a. Devices shall be handled on benches with conductive handling devices.

b. Ground test equipment, tools, and personnel handling devices.

c. Do not handle devices by the leads.

d. Store devices in conductive foam or carriers.

e. Avoid use of plastic, rubber, or silk in MOS areas.

f. Maintain relative humidity above 50 percent, if practical.

g. Care shall be exercised, during test and troubleshooting, to apply not more than maximum rated voltage to any lead.

h. Gate must be terminated to source. $R \leq 100$ k, whenever bias voltage is to be applied drain to source.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-S-19500 and table II herein. Alternate flow is allowed for qualification inspection in accordance with figure 2 of MIL-S-19500.

4.2.1 Group E inspection. Group E inspection shall be conducted in accordance with MIL-S-19500 and table II herein.

4.2.2 JANHC and JANKC die. Qualification shall be in accordance with MIL-S-19500.

4.3 Screening (JANS, JANTX, and JANTXV levels only). Screening shall be in accordance with table II of MIL-S-19500 and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

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4.3.1 Screening (JANHC and JANKC). Screening of JANC die shall be in accordance with MIL-S-19500, appendix H. As a minimum, die shall be 100-percent probed in accordance with group A, subgroup 2. The maximum test current shall be 20 A dc.

Screen (see table II of MIL-STD-19500)	Measurement	
	JANS level	JANTX and JANTXV levels
1/	Gate stress test (see 4.5.5)	Gate stress test (see 4.5.5)
1/	Method 3470 (see 4.5.4)	Method 3470 (see 4.5.4)
1/	Method 3161 (see 4.5.3)	Method 3161 (see 4.5.3)
3	Test condition G	Test condition G
9 1/	I_{GSS1}, I_{DSS1}, subgroup 2 of table I herein.	Subgroup 2 of table I herein
10	Method 1042, test condition B	Method 1042, test condition B
11	I_{GSS1}, I_{DSS1}, $r_{DS(on)1}$, $V_{GS(th)1}$, subgroup 2 of table I herein; ΔI_{GSS1} = ±20 nA dc or ±100 percent of initial value, whichever is greater. ΔI_{DSS1} = ±25 μA dc or ±100 percent of initial value, whichever is greater.	I_{GSS1}, I_{DSS1}, $r_{DS(on)1}$, $V_{GS(th)1}$, subgroup 2 of table I herein.
12	Method 1042, test condition A, t = 240 hours	Method 1042, test condition A
13	Subgroups 2 and 3 of table I herein; ΔI_{GSS1} = ±20 nA dc or ±100 percent of initial value, whichever is greater. ΔI_{DSS1} = ±25 μA dc or ±100 percent of initial value, whichever is greater. $\Delta r_{DS(on)1}$ = ±20 percent of initial value. $\Delta V_{GS(th)1}$ = ±20 percent of initial value.	Subgroup 2 of table I herein; ΔI_{GSS1} = ±20 nA dc or ±100 percent of initial value, whichever is greater. ΔI_{DSS1} = ±25 μA dc or ±100 percent of initial value, whichever is greater. $\Delta r_{DS(on)1}$ = ±20 percent of initial value. $\Delta V_{GS(th)1}$ = ±20 percent of initial value.

1/ Shall be performed anytime before screen 10.

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-S-19500. Alternate flow is allowed for quality conformance inspection in accordance with figure 2 of MIL-S-19500.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-S-19500 and table I herein. Electrical measurements (end-points) and delta requirements shall be in accordance with the applicable of table I, group A, subgroup 2 herein.

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4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table IVa (JANS) and table IVb (JAN, JANTX, and JANTXV) of MIL-S-19500 and herein. Electrical measurements (end-points) and delta requirements shall be in accordance with the applicable of table I, group A, subgroup 2 herein.

4.4.2.1 Group B inspection, table IVa (JANS) of MIL-S-19500.

Subgroup	Method	Condition
3	1051	Test condition G.
4	1042	Test condition D; 2,000 cycles. The heating cycle shall be 1 minute minimum.
5	1042	Accelerated steady-state operation life; method 1042; test condition A, V_{DS} = rated, T_A = +175°C, t = 120 hours. Read and record $V_{(BR)DSS}$ (pre and post) at 1 mA = I_D. Read and record I_{DSS} (pre and post) in accordance with table 1, group A, subgroup 2. Delta $V_{BR(DSS)}$ cannot exceed 10 percent; delta I_{DSS} cannot exceed 50 μA. Accelerated steady-state gate stress; condition B, V_{GS} = rated, T_A = +175°C, t = 24 hours.
6	3161	See 4.5.2.

4.4.2.2 Group B inspection, table IVb (JAN, JANTX, and JANTXV) of MIL-S-19500.

Subgroup	Method	Condition
2	1051	Test condition G.
3	1042	Test condition D, 2,000 cycles. The heating cycle shall be 1 minute minimum.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table V of MIL-S-19500 and as follows. Electrical measurements (end-points) and delta requirements shall be in accordance with the applicable of table I, group A, subgroup 2 herein.

Subgroup	Method	Condition
2	2036	Test condition A, weight = 10 pounds, t = 10 seconds.
6	1042	Test condition D, 6,000 cycles. The heating cycle shall be 1 minute minimum.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

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4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

4.5.2 Thermal resistance. Thermal resistance measurements shall be performed in accordance with method 3161 of MIL-STD-750. $R_{\theta JC(max)}$ = 0.50°C/W.

a.	Measuring current (I_M)	10 mA.
b.	Drain heating current (I_H)	8 A.
c.	Heating time (t_H)	Steady-state (see MIL-STD-750, method 3161 for definition).
d.	Drain-source heating voltage (V_H)	25 V dc.
e.	Measurement time delay (t_{MD})	30 to 60 μs.
f.	Sample window time (t_{SW})	10 μs maximum.

4.5.3 Thermal impedance ($Z_{\theta JC}$ measurements). The $Z_{\theta JC}$ measurements shall be performed in accordance with MIL-STD-750, method 3161. The maximum limit (not to exceed figure 2, thermal impedance cures and the group A, subgroup 2 limits) for $Z_{\theta JC}$ in screening (see table II of MIL-S-19500) shall be derived by each vendor by means of statistical process control. When the process has exhibited control and capability, the capability data shall be used to establish the fixed screening limit. In addition to screening, once a fixed limit has been established, monitor all future sealing lots using a random five piece sample from each lot to be plotted on the applicable X, R chart. If a lot exhibits an out of control condition, the entire lot shall be removed from the line and held for Engineering evaluation and disposition. This procedure may be used in lieu of an in line process monitor.

a.	Measuring current (I_M)	10 mA.
b.	Drain heating current (I_H)	8 A minimum.
c.	Heating time (t_H)	100 ms.
d.	Drain-source heating voltage (V_H)	25 V dc minimum.
e.	Measurement time delay (t_{MD})	30 to 60 μs.
f.	t_{SW} sample window time	10 μs (maximum).

4.5.4 Single pulse avalanche energy (E_{AS}).

a.	Peak current (I_{AS})	$I_{AS(max)}$.
b.	Peak gate voltage (V_{GS})	10 V dc.
c.	Gate to source resistor (R_{GS})	25 Ω ≤ R_{GS} ≤ 200 Ω.
d.	Initial case temperature (T_C)	+25°C +10°C, -5°C.
e.	Inductance (L)	$[2E_{AS}/(I_{D1})^2]$ $\frac{[(V_{BR} - V_{DD})]}{V_{BR}}$ mH minimum
f.	Number of pulses to be applied	1 pulse minimum.
g.	Supply voltage (V_{DD})	50 V dc.

4.5.5 Gate stress test.

V_{GS} = ±30 V dc minimum.

t = 250 μs minimum.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical inspection	2071					
Subgroup 2						
Thermal impedance 2/	3161	See 4.5.3	$Z_{\theta JC}$		0.4	°C/W
Breakdown voltage, drain to source	3407	V_{GS} = 0 V dc; I_D = 1.0 mA dc; bias condition C	$V_{(BR)DSS}$			
2N7229				400		V dc
2N7230				500	→	V dc
Gate to source voltage (threshold)	3403	$V_{DS} \geq V_{GS}$; I_D = 0.25 mA dc	$V_{GS(th)1}$	2.0	4.0	V dc
Gate current	3411	Bias condition C; V_{DS} = 0 V dc; V_{GS} = +20 V dc and -20 V dc	I_{GSS1}		±100	nA dc
Drain current	3413	V_{GS} = 0; bias condition C; V_{DS} = 0 V dc; V_{DS} = 80 percent of rated V_{DS}	I_{DSS1}		25	µA dc
Static drain to source on-state resistance	3421	V_{GS} = 10 V dc; condition A; pulsed (see 4.5.1); I_D = I_{D2}	$r_{DS(on)1}$			
2N7229					0.200	Ω
2N7230					0.270	Ω
Drain to source on-state resistance	3421	V_{GS} = +10 V dc; condition A; pulsed (see 4.5.1); I_D = I_{D1}	$r_{DS(on)2}$			
2N7229					0.23	Ω
2N7230					0.31	Ω
Forward voltage (source drain diode)	4011	Pulsed (see 4.5.1); V_{GS} = 0 V dc; I_S = I_{D1}	V_{SD}		1.8	
Subgroup 3						
High temperature operation:		T_C = T_J = +125°C				
Gate current	3411	Bias condition C; V_{DS} = 0 V dc; V_{GS} = +20 V dc and -20 V dc	I_{GSS2}		±200	nA dc

See footnote at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 3 - Continued						
Drain current	3413	Bias condition C; V_{GS} = 0 V dc; V_{DS} = 80 percent rated V_{DS}	I_{DSS3}		0.25	mA dc
Static drain to source on-state resistance	3421	V_{GS} = 10 V dc; I_D = I_{D2}; pulsed (see 4.5.1)	$r_{DS(on)3}$			
2N7229					0.395	Ω
2N7230					0.525	Ω
Gate to source voltage (threshold)	3403	$V_{DS} \geq V_{GS}$; I_D = 0.25 mA	$V_{GS(th)2}$	1.0	→ 5	V dc
Low temperature operation:		T_C = T_J = -55°C				
Gate to source voltage (threshold)	3403	$V_{DS} \geq V_{GS}$; I_D = 0.25 mA	$V_{GS(th)3}$		5.0	V dc
Subgroup 4						
Switching time test	3472	I_D = rated I_{D2} (see 1.3); V_{GS} = 10 V dc; R_g = 2.35 Ω; V_{DD} = 0.5 $V_{BR(DSS)}$				
Turn-on delay time			$t_{d(on)}$		40	nS
Rise time			t_r		140	nS
Turn-off delay time			$t_{d(off)}$		130	nS
Fall time			t_f		100	nS
Subgroup 5						
Safe operating area test	3474	See figures 3 and 4; V_{DS} = 80 percent of rated V_{DS}; V_{DS} = 200 V dc maximum; t_p = 10 ms				
2N7229		I_D = 5.0 A				
2N7230		I_D = 3.4 A				
Electrical measurements		See table I, group A, subgroup 2				

See footnote at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 6						
Not applicable						
Subgroup 7						
Gate charge	3471	Condition B				
Test 1						
On-state gate charge			$Q_{g(on)}$			nC
2N7229					210	
2N7230					200	
Test 2						
Gate to source charge			Q_{gs}			nC
2N7229					35	
2N7230					35	
Test 3						
Gate to drain charge			Q_{gd}			nC
2N7229					120	
2N7230					135	
Reverse recovery time	3473	$V_{DD} \leq +50$ V dc	t_{rr}			
2N7229		$di/dt \leq +100$ A/µs; $I_F = I_{D1}$ A;			1,000	ns
2N7230		$di/dt \leq +100$ A/µs; $I_F = I_{D1}$ A			1,200	ns

1/ For sampling plan, see MIL-S-19500.

2/ This test is required for the following end-point measurements only (not intended for screen 13):

JANS - group B, subgroups 3 and 4

JAN, JANTX, and JANTXV - group B, subgroups 2 and 3;
group C, subgroup 6;
group E, subgroup 1

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TABLE II. Group E inspection (all quality levels) for qualification only.

Inspection 1/	MIL-STD-750		Qualification conformance inspection
	Method	Conditions	
Subgroup 1			45 devices, c = 0
Thermal shock (temperature cycling)	1051	Condition G, 500 cycles	
Electrical measurements		See table I, group A, subgroup 2	
Subgroup 2 2/			45 devices, c = 0
Steady-state reverse bias	1042	Condition A; 1,000 hours	
Electrical measurements		See table I, group A, subgroup 2	
Steady-state gate bias	1042	Condition B; 1,000 hours	
Electrical measurements		See table I, group A, subgroup 2	
Subgroup 3			
Not applicable			
Subgroup 4			5 devices, c = 0
Thermal resistance	3161	$R_{\theta JC}$ = 0.50°C/W maximum (see 4.5.2)	
Subgroup 5			
Barometric pressure (reduced)	1001	Test condition C; $V_{ISO} = V_{DS}$; I_{ISO} 0.25 mA maximum	

1/ JANHC and JANKC devices are qualified in accordance with appendix H of MIL-S-19500.
2/ A separate sample may be pulled for each test.

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2N7229 AND 2N7230

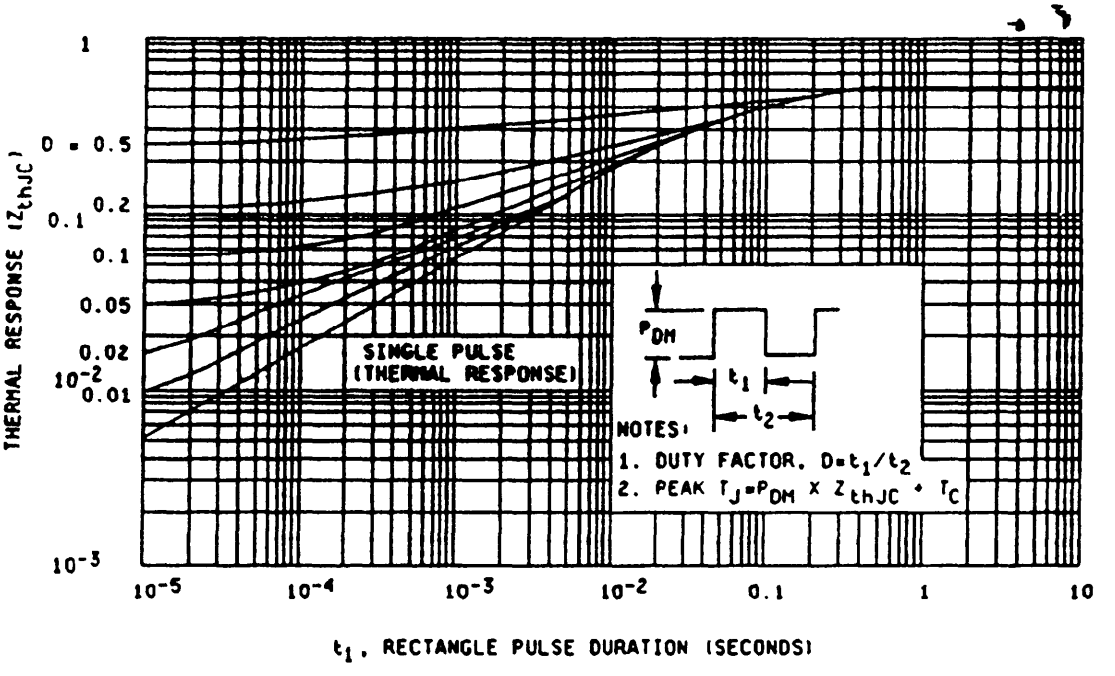


FIGURE 3. Transient thermal response.

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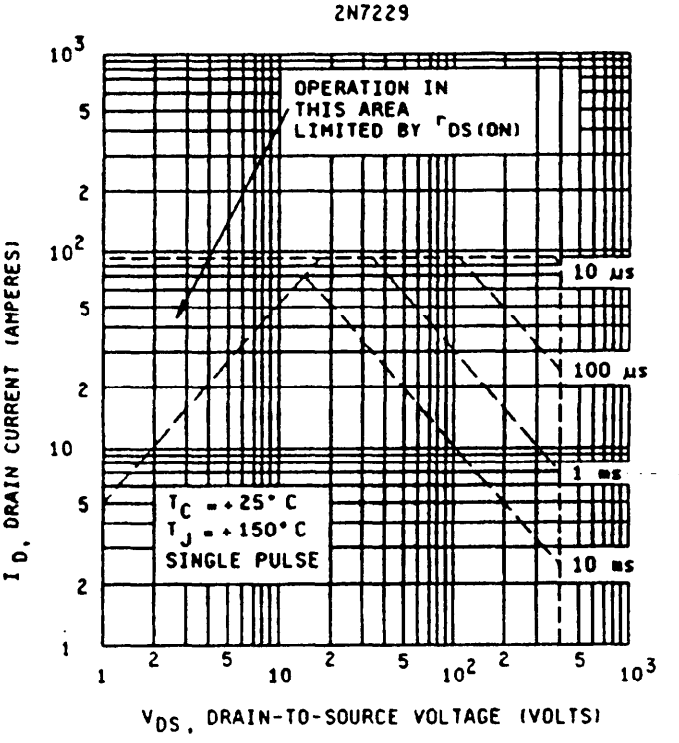


FIGURE 4. Maximum safe operating area.

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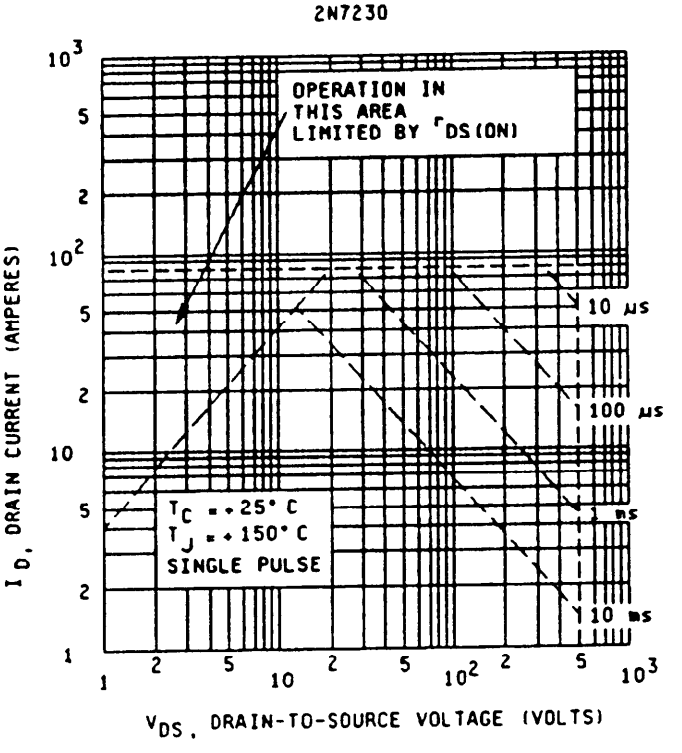


FIGURE 4. Maximum safe operating area - Continued.

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5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-S-19500.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Issue of DODISS to be cited in the solicitation (see 2.1.1 and 2.2).

b. The lead finish and formation as specified (see 3.3.1).

c. Product assurance level, type designation and for die acquisition manufactures letter identification.

6.3 Suppliers of die. The qualified die suppliers will be identified on the QPL (example, JANHCA2N7229).

JANC ordering information	
	Manufacturer
PIN	59993
2N7229	JANHCA2N7229
2N7230	JANHCA2N7230

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Navy - TD
Air Force - 19, 70, 80

Preparing activity:
DLA - ES

(Project 5961-1767)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-S-19500/600A	2. DOCUMENT DATE (YYMMDD) 95/04/07

3. DOCUMENT TITLE
SEMICONDUCTOR DEVICE, FIELD EFFECT TRANSISTOR, N-CHANNEL, SILICON TYPES 2N7229 AND 2N7230 JAN, JANTX, JANTXV, JANS, JANHC, AND JANKC

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME Alan Barone	b. TELEPHONE (Include Area Code) (1) Commercial 513-296-6048 (2) AUTOVON 986-6048
c. ADDRESS (Include Zip Code) Defense Electronics Supply Center Attn: DESC-ELDT Dayton, Ohio 45444-5765	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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